

MARKNESSE, Netherlands

WMO: 062730

Lat: 52.7019N

Lon: 5.8875E

Elev: -10

StdP: 14.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.2	(c) 23.8	(d) 14.6	(e) 11.5	(f) 23.3	(g) 19.2	(h) 14.4	(i) 24.5	(j) 29.8	(k) 46.5	(l) 26.9	(m) 46.7	(n) 8.3	(o) 80	(p) 0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.6	(c) 82.3	(d) 68.0	(e) 78.4	(f) 66.1	(g) 74.9	(h) 64.6	(i) 70.0	(j) 79.5	(k) 68.1	(l) 75.7	(m) 66.2	(n) 73.0	(o) 8.8	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	98.6	74.0	65.1	92.7	71.7	63.5	87.6	69.7	33.9	79.4	32.3	76.0	30.9	73.1	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.1	(b) 21.2	(c) 18.9	(e) 15.7	(f) 90.4	(g) 8.8	(h) 3.2	(i) 9.4	(j) 92.7	(k) 4.3	(l) 94.6	(m) -0.7	(n) 96.5	(o) -7.1	(p) 98.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			15.0	74.8	8.6	2.0	8.8	76.2	3.7	77.4	-1.1	78.6	-7.4	80.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	37.3	38.6	42.9	49.5	55.9	60.3	63.7	63.2	58.6	51.4	44.4	39.0
	DBStd	11.16	7.16	6.65	5.95	6.34	6.10	5.05	5.15	4.44	4.68	5.81	5.81	7.37
	HDD50	1616	393	321	226	85	13	0	0	0	1	54	181	342
	HDD65	5415	858	740	685	465	289	160	88	87	199	421	618	805
	CDD50	1792	1	1	7	71	197	308	424	411	259	97	13	3
	CDD65	114	0	0	0	0	8	18	48	33	7	0	0	0
	CDH74	946	0	0	0	16	95	168	388	229	49	1	0	0
	CDH80	221	0	0	0	1	12	40	107	54	7	0	0	0
Wind	WSAvg	9.4	11.3	10.8	10.4	9.5	9.3	8.2	7.7	7.5	7.8	9.3	9.7	10.9
Precipitation	PrecAvg	34.3	2.9	2.4	2.1	1.8	2.5	3.1	3.7	3.6	3.1	3.0	3.0	3.0
	PrecMax	46.3	5.5	5.4	4.2	3.9	5.0	6.8	7.2	7.9	7.4	9.4	5.8	5.9
	PrecMin	25.8	0.3	0.8	0.4	0.0	1.1	1.3	1.6	0.4	0.4	0.7	0.4	1.0
	PrecStd	4.7	1.4	1.2	1.2	0.9	1.0	1.4	1.6	1.9	1.7	1.6	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	55.2	63.1	76.2	81.6	85.1	88.7	85.8	79.7	70.0	60.3	55.3
		MCWB	51.6	49.4	53.7	59.6	64.7	70.1	69.6	68.9	67.2	62.9	56.7	53.1
	2%	DB	51.0	51.7	58.4	69.6	76.3	78.7	82.7	80.1	74.2	64.5	57.0	53.0
		MCWB	49.0	48.6	51.3	56.5	60.8	66.6	68.5	68.2	65.4	60.0	54.8	51.1
	5%	DB	48.6	49.4	54.0	64.8	72.0	74.0	78.2	75.5	70.1	61.8	54.7	51.0
		MCWB	46.8	46.9	49.6	54.8	60.0	64.5	67.0	65.9	63.0	58.1	52.5	49.2
	10%	DB	46.7	47.2	51.2	60.4	67.5	70.2	74.2	72.2	66.8	59.5	52.4	48.4
		MCWB	45.0	45.2	47.8	52.9	58.2	62.4	65.4	64.0	61.2	56.6	50.5	46.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	51.2	55.2	61.2	67.4	71.9	72.6	72.3	68.9	63.4	57.3	53.6
		MCDB	53.5	53.7	60.2	72.6	76.9	81.9	82.8	81.9	77.1	68.6	59.1	54.8
	2%	WB	49.3	49.0	52.2	58.1	63.9	68.4	70.1	69.0	66.0	60.7	54.9	51.5
		MCDB	50.8	51.0	56.2	67.1	72.4	76.1	80.1	77.0	72.8	64.1	56.8	52.9
	5%	WB	47.1	47.1	50.2	55.7	61.2	65.8	68.3	67.0	63.9	58.8	52.8	49.3
		MCDB	48.5	49.1	53.8	63.2	68.7	72.6	76.5	73.7	68.8	61.3	54.4	50.8
	10%	WB	45.2	45.4	48.2	53.7	59.0	63.4	66.3	65.2	62.2	57.1	50.7	47.0
		MCDB	46.5	47.2	51.1	59.5	66.2	68.9	72.8	70.6	66.1	59.3	52.2	48.4
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.7	12.0	15.9	15.8	15.5	15.6	15.4	14.3	11.8	8.6	7.0
		MCDBR	7.7	11.3	17.8	23.6	23.8	22.2	23.4	21.4	19.5	13.6	9.6	7.7
		MCWBR	7.2	8.9	12.1	12.7	11.4	12.3	11.4	11.1	11.3	9.4	8.0	7.2
	5% WB	MCDBR	7.8	10.3	15.8	20.5	19.7	20.1	20.8	19.0	17.3	13.2	9.4	7.8
MCWBR		7.5	8.7	11.3	11.7	10.8	12.1	11.2	10.8	10.6	9.8	8.2	7.4	
Clear-Sky Solar Irradiance	taub	0.332	0.352	0.381	0.407	0.409	0.406	0.418	0.411	0.389	0.366	0.344	0.325	
	taud	2.317	2.320	2.254	2.233	2.270	2.294	2.271	2.310	2.361	2.386	2.343	2.311	
	Ebn at Noon	204	237	253	261	267	269	263	258	249	229	196	182	
	Edh at Noon	20	26	34	39	40	39	40	36	31	24	19	17	
All-Sky Solar Radiation	RadAvg	207	421	810	1318	1618	1708	1632	1360	971	551	244	157	
	RadStd	21	73	91	129	114	115	178	108	92	51	23	17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page